

Some calculations with acceleration

Two blindfolded passengers are sat in a car. The vehicle brakes heavily to stop at a red light. One remarks “I felt a large acceleration”. The other says “Are you mad? It was obviously heavy braking”.

- a. The first passenger is a mathematician. Explain what he means by *acceleration*. What does he mean here by “a large acceleration”?
- b. An accelerometer in the car registered a constant reading of -7 ms^{-2} (meaning *metres per second per second*) during the braking. Alternative units of acceleration are miles per hour per second. Convert the reading into these units. [*There are 1609 metres in one mile.*]
- c. The truck was braking from 60 mph. How long did it take to stop? What was its average velocity (in miles per hour) during the braking? How much distance did it cover?